



BITS Pilani presentation

BITS Pilani
Pilani Campus

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SS ZG622

Software Project Management

Lecture No. 5

Recap - Function point analysis



- [Suggested solution](#)
- A realistic estimate that includes all activities
- Possible repetition of certain points
- Detailed FPA estimation once the design is completed

Planning in agile projects



- Overview of agile project planning
- Product backlog (evolving system scope)
- Prioritizing and sprint backlog
- Estimating story points
- Burndown charts for project management
- Decomposition of complex user stories

Overview of Agile Projects

Planning



- Iterative, adaptive and collaborative process
- Spread throughout the project life cycle
- Product vision and roadmap
- Release planning and sprint planning
- Planning during sprint review and retrospective

Activity W05#1



Write user stories for the Mux Core Sales system
(you may refer to [these functional specifications](#))

Please enter your set of user stories [here](#)
(link shared in Chat panel)

Activity W04#2



Prioritize the given user stories by
changing the order of the stories
(high value ones at the top)



<https://pollev.com/narsibolloju019>

Scrum - estimation



- Traditional software teams give estimates in a time format: days, weeks, months.
- Many agile teams, however, have transitioned to story points. Story points rate the relative effort of work using
 - Fibonacci-like format: 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, ...
 - Powers of 2: 1, 2, 4, 8, 16, 32, 64, ...
 - T-shirt sizes: XXS, XS, S, M, L, XL, XXL
 - ...



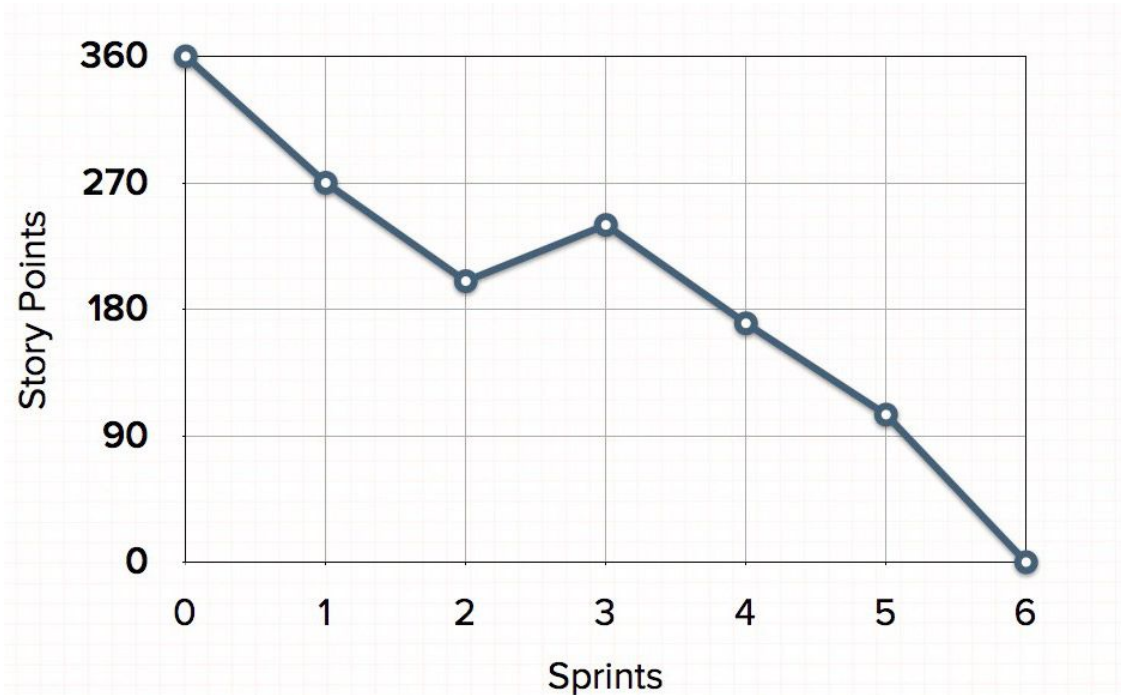
Planning Poker – estimation technique



- All participants use numbered playing cards and estimate the items
- Individual estimates are done and discussion is raised when there are large differences
- This process is repeated till the whole team reached consensus about the agreed estimation (which is not an average of estimates!)

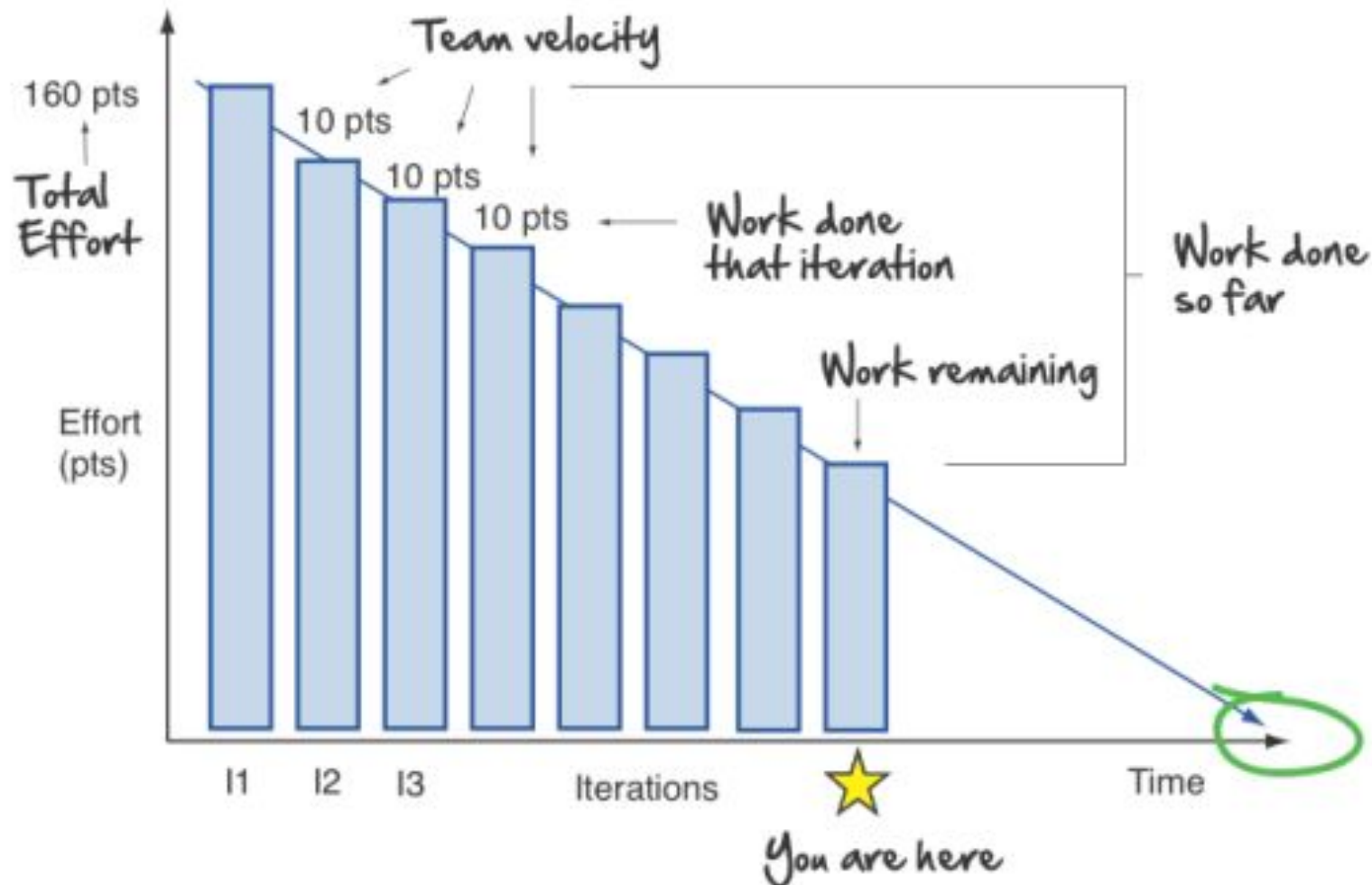
<https://technology.amis.nl/2016/03/23/8-agile-estimation-techniques-beyond-planning-poker/>

Burndown Chart - example



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Burndown Charts



Activity W05#3



Decompose the user story
“sell movie tickets to customers”
into several simpler user stories

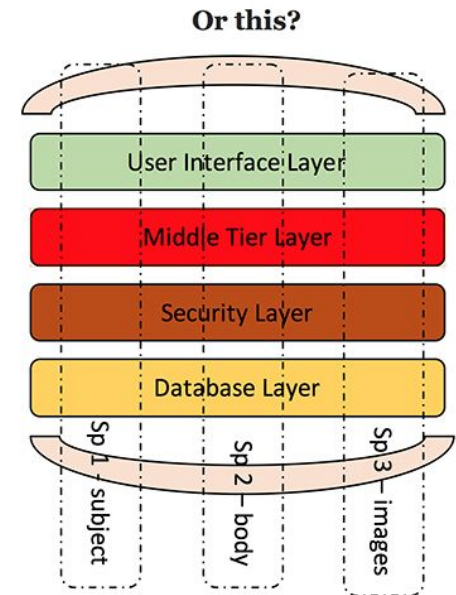
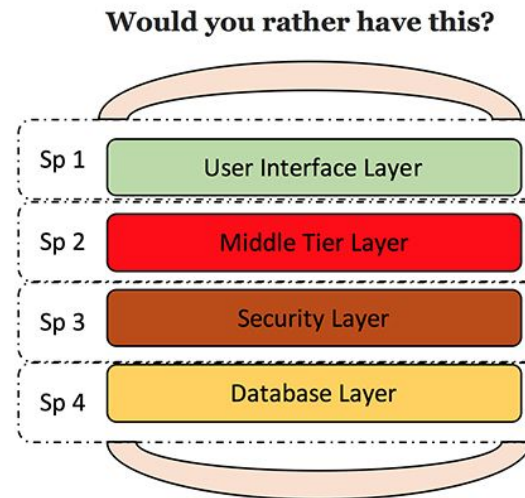
User story decomposition - horizontal vs. vertical slicing



<https://gojko.net/2012/01/23/splitting-user-stories-the-hamburger-method/>

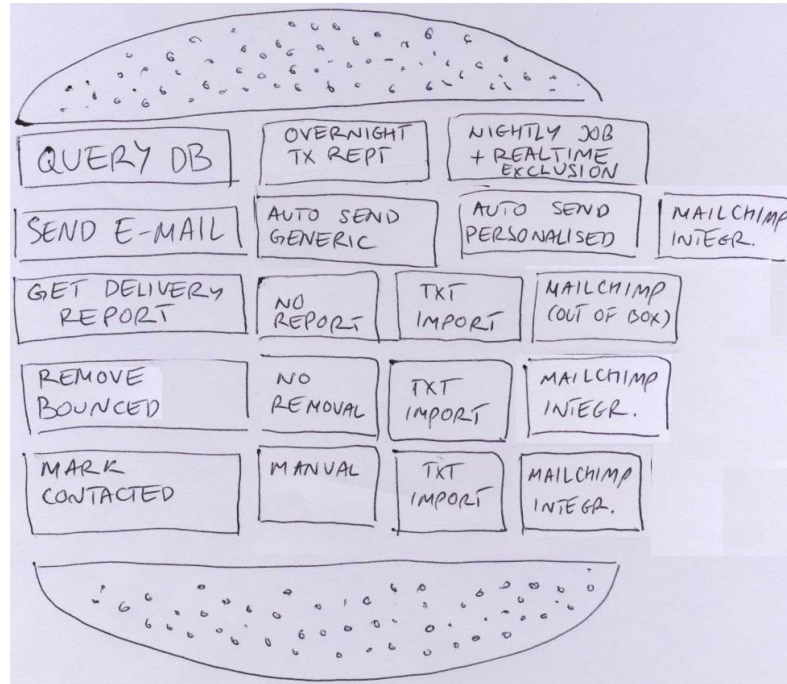
Splitting user stories - vertical vs. horizontal slicing

- Vertical slicing for delivering functional piece of the story that delivers value to the end user
- Horizontal slicing breaks a complex story into smaller parts (but do not deliver value to the end user until all the parts are completed)



<https://www.apple-brook.com/vertically-slicing-taking-a-bite-of-the-hamburger/>

User story decomposition - vertical slicing



<https://gojko.net/2012/01/23/splitting-user-stories-the-hamburger-method/>

Splitting user stories - The hamburger method



1. Identify tasks
2. Identify options for tasks
3. Combine results
4. Trim the hamburger
5. Take the first bite
6. Take another bite ... and continue

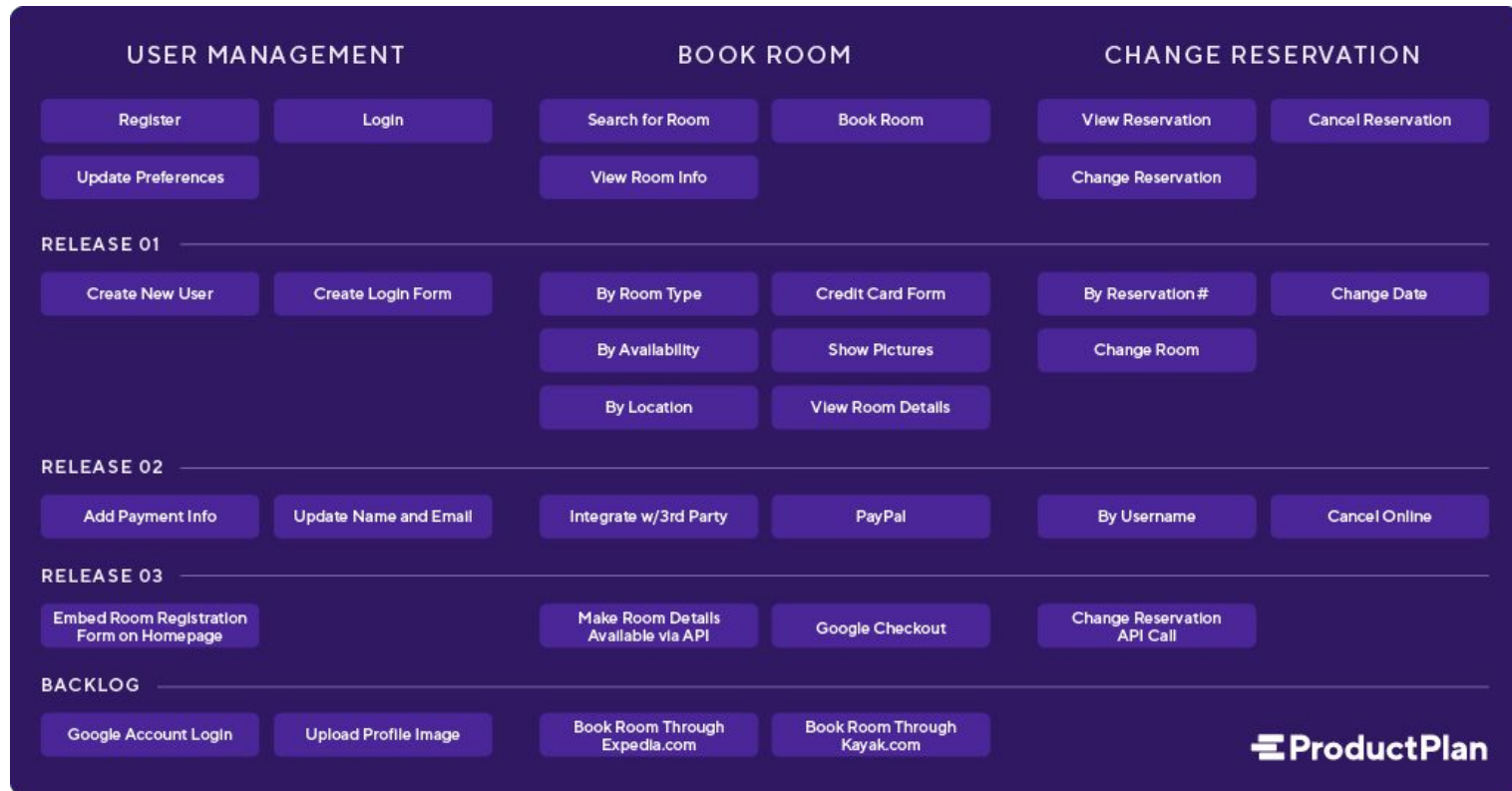
<https://gojko.net/2012/01/23/splitting-user-stories-the-hamburger-method/>

Activity W05#4



Decompose the user story
“sell movie tickets to customers”
into several simpler user stories
using the Hamburger method

Story mapping technique



Activity W05#5



Compare and contrast project planning between
plan-driven and agile models

Additional slides



Definition of Ready (for a user story)

- The story is defined
- The story dependencies are identified
- The story is sized by the team
- Team accepts user experience artefacts
- The story has a **well-defined acceptance criteria** (including performance criteria where appropriate)
- Team has a good idea what it will mean to demo the story
- Person who will accept the story is identified

<https://www.quickscrum.com/ScrumGuide/169/sg-Definition-Of-Ready>

A Good User Story Is

Independent (of all others)

Negotiable (not a contract for a specific feature)

Valuable (a vertical slice)

Estimable (to a decent approximation)

Small (enough to fit in a single iteration)

Testable (even if the test doesn't exist yet)

<https://www.knowledgehut.com/tutorials/scrum-tutorial/definition-of-ready>

Acceptance Criteria

- Acceptance criteria are a set of statements, each with a clear pass/fail result, that specify both functional and non-functional requirements.
- These requirements represent “conditions of satisfaction.”
- There is no partial acceptance: **either a criterion is met or it is not.**

<http://www.seguetech.com/what-characteristics-make-good-agile-acceptance-criteria/>

Scenario: Handling loyalty card during sale

Given a customer presents a loyalty card and the staff scans/verifies it in the system

When the staff applies any discounts or points redemption to the ticket price and completes the purchase

Then the system updates the loyalty points balance, reflects the discounted total on the ticket, and sends a confirmation SMS/email if opted in.

GWT specification of acceptance criteria (Format 1) example



Scenario	Given	When	Then
User wants to log in to his/her registered account	User is on the login page	User fills in correct username AND fills in correct password AND clicks the login button	System presents the dashboard page
	User is on the login page	User fills in correct username AND fills in wrong password AND clicks the login button	System informs the user that the password is wrong
	User is on the login page	User clicks on forgot password link	System displays password change request page

GWT specification of acceptance criteria (Format 2) example



Scenario	Given	When	Then
User wants to reset password of his/her registered account	User submits forgot password form with an email address	The email address is invalid	System informs that email address is not valid
	User submits forgot password form with an email address	The email address is valid AND the email address is not registered	System informs that email address is not registered
	User submits forgot password form with an email address	The email address is valid AND the email address is registered	System sends an email to that address with instructions for resetting the password or disregarding the email message